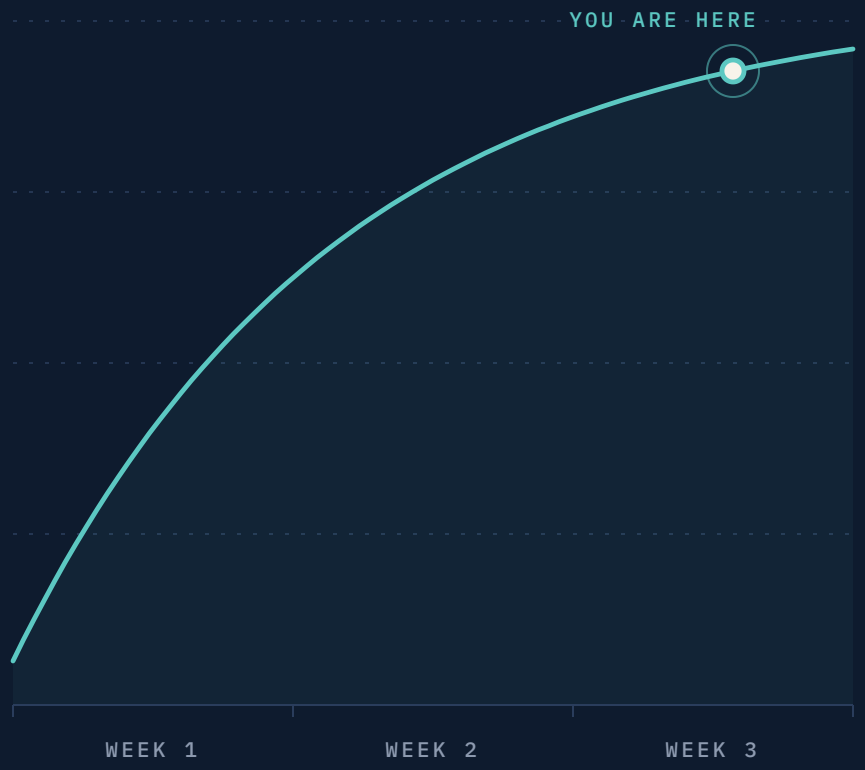
WEEK 3 · DAY 3 · OVERVIEW

Resampling, SMOTE, and Class Weights



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ABOUT 50 MIN WITH EXERCISES / 11 SECTIONS / 4 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Problem Statement

BUSINESS CONTEXT

Of the 4,680 Polish companies in the training rows, 326 went bankrupt within a year: about 13 healthy companies for every bankrupt one.

Tutorials and AI answers give the same advice for a file like this: before you fit, weight the rare class, copy it, drop some of the common class, or make up new rare rows.

Course 4's capstone took the first route, and its logistic regression carries `class_weight="balanced"`.

Day 12 handed over two models: LightGBM at its defaults, at a fold average precision of 0.7783, and that logistic pipeline, at 0.6624.

The colleague who brought the file asks three questions: does a fix help either model rank, what does it do to the output, and can the output still be read as a probability?

The companies date from 2000 to 2013 (dataset page): one country, one period.

An unweighted model's output is the model's probability of bankruptcy in this file; a weighted or resampled model's output is the model's score.

OBJECTIVES

Weight the rare class with class weights, resample the rows three ways and count what each leaves, and build one synthetic row by hand.

Score weighted, resampled and unweighted models on the same folds and at one recall, measure what each fix does to a model's output, and turn a score back into a probability two ways.

ABOUT 50 MIN WITH EXERCISES / 11 SECTIONS / 4 EXERCISES

Your Route Through Day 3

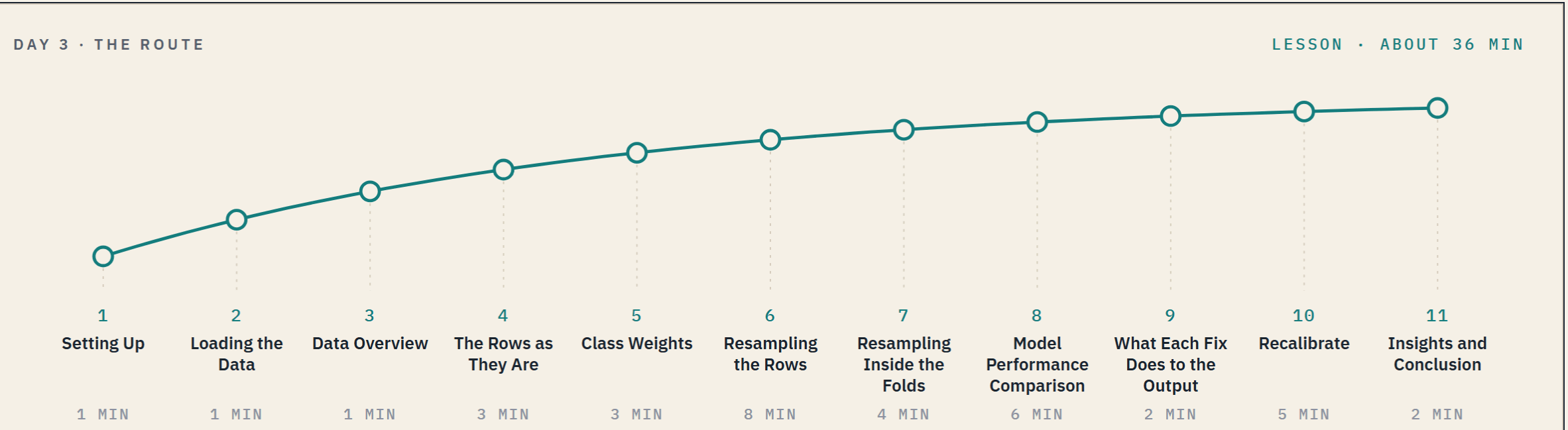


FIGURE 1 Eleven sections, in order, with about how long each takes.

Time. About 36 minutes for the lesson, then about 12 for the four exercises.

Fix the seed and load the libraries.

Setting Up

THE SETUP

```
import os
from pathlib import Path
os.environ["OMP_NUM_THREADS"] = "1"
os.environ["OPENBLAS_NUM_THREADS"] = "1"
SEED = 42
DATA = "../../../data/"
```

TODAY'S TOOLS

```
from imblearn.over_sampling import SMOTE, RandomOverSampler
from imblearn.under_sampling import RandomUnderSampler
from imblearn.pipeline import Pipeline as ImbPipeline
from sklearn.neighbors import NearestNeighbors
from sklearn.calibration import CalibratedClassifierCV, calibration_curve
```

THE TWO RATIOS THIS NOTEBOOK NAMES

- **`Attr1`** to **`Attr64`**: the page's X1 to X64, ratios from the statements
- **`class`**: 1 if the company went bankrupt within a year, **the target**
- **`Attr2`**: total liabilities / total assets
- **`Attr7`**: EBIT / total assets

Run this cell and move on: it fixes the seed and finds the course data.

Loading the Data

COURSE 4'S ROWS

```
bankrupt = pd.read_csv(DATA + "polish_bankruptcy_5year.csv")
unique = bankrupt.drop_duplicates()
X_train, X_test, y_train, y_test = train_test_split(unique[FEATURES], unique[TARGET], test_size=0.2,
print(len(bankrupt), "read,", len(unique), "unique.  Training:", len(X_train), "rows,", y_train.sum(), "bankrupt")
```

```
5910 read, 5850 unique.  Training: 4680 rows, 326 bankrupt
```

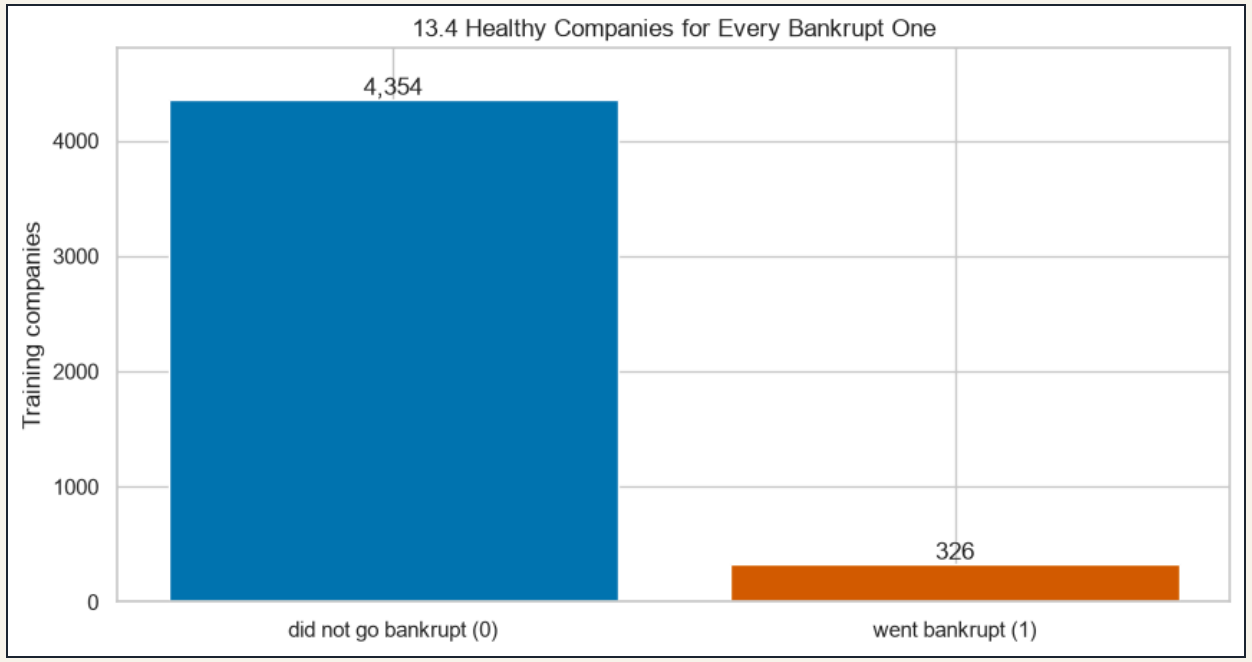
DAY 11'S FOLDS

```
folds = StratifiedKFold(n_splits=5, shuffle=True, random_state=SEED)
went = y_train.to_numpy() == 1
```

The same 4,680 training rows as Course 4's capstone and Days 11 and 12.

Count how rare bankruptcy is in training.

Data Overview



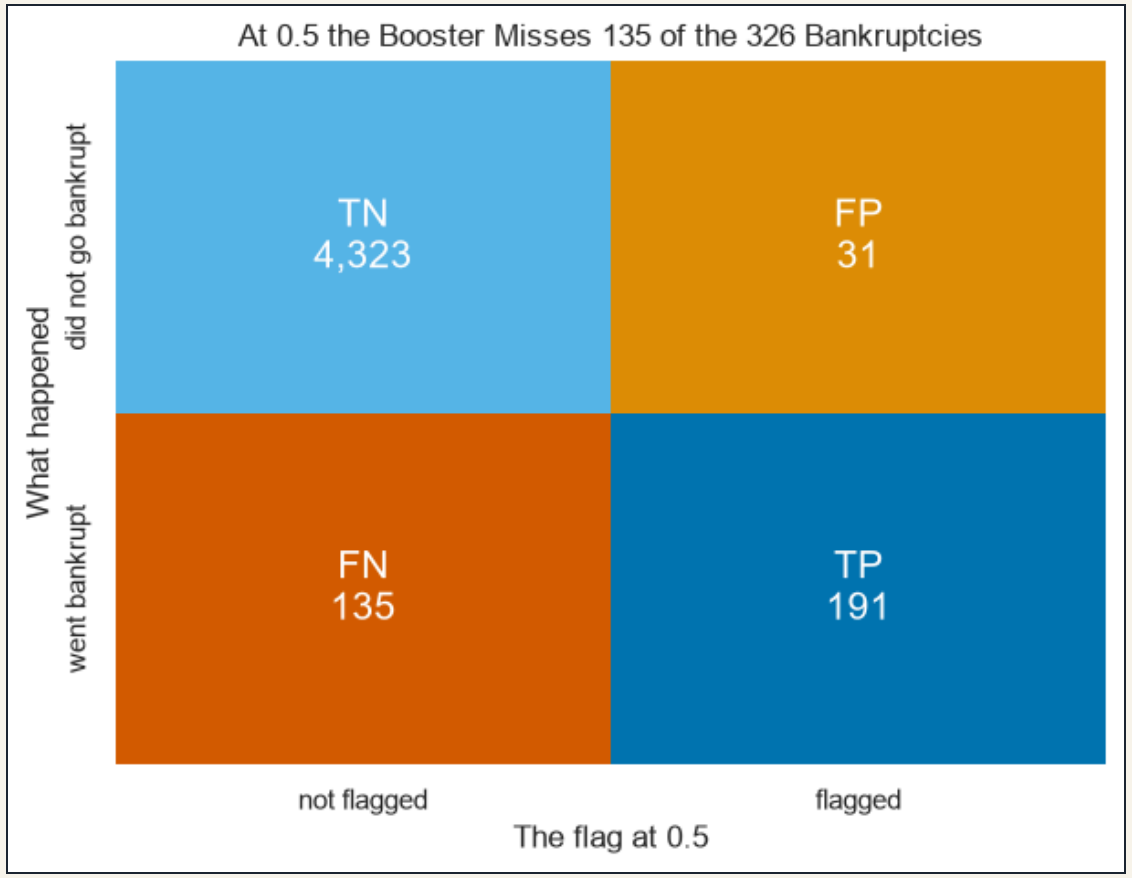
HOW RARE IS RARE

```
counts = y_train.value_counts()

{0: 4354, 1: 326}  Healthy per bankrupt: 13.4
Right if nobody is flagged: 0.9303  Bankrupt share: 0.0697
```

Average precision ranks the models, and recall at a stated threshold counts the catches, as all week.

The Rows as They Are



THE BOOSTER AT ITS DEFAULTS

```
booster = LGBMClassifier(random_state=SEED, n_jobs=1, verbose=-1)

Folds: [0.7077 0.7988 0.7434 0.8013 0.8405]  Mean: 0.7783
```

THE FLAGS AT 0.5

```
flags = plain >= 0.5

Flagged: 222  Caught: 191 of 326  Missed: 135
```

At the default cut the model misses four bankruptcies in ten: the problem the advice promises to fix.

Class Weights

WHAT BALANCED WEIGHTS ARE

```
weights = len(y_train) / (2 * counts)
```

```
Weights: {0: 0.5374, 1: 7.1779}  Totals: {0: 2340.0, 1: 2340.0}
```

THE QUESTION

Q. Take the weights out and change nothing else. Write down its fold average precision.

COURSE 4'S PIPELINE, WITH OR WITHOUT CLASS WEIGHTS

```
def logistic(weights=None):
    return Pipeline([("impute", SimpleImputer(strategy="median", add_indicator=True)),
                     ("quantile", QuantileTransformer(output_distribution="normal", random_state=SEED)),
                     ("logit", LogisticRegression(solver="newton-cholesky", tol=1e-8, class_weight=weights))])
course4 = out_of_fold(logistic("balanced"))
```

Each bankrupt company counts 7.18 times, each healthy one about half a time.

Resampling the Rows

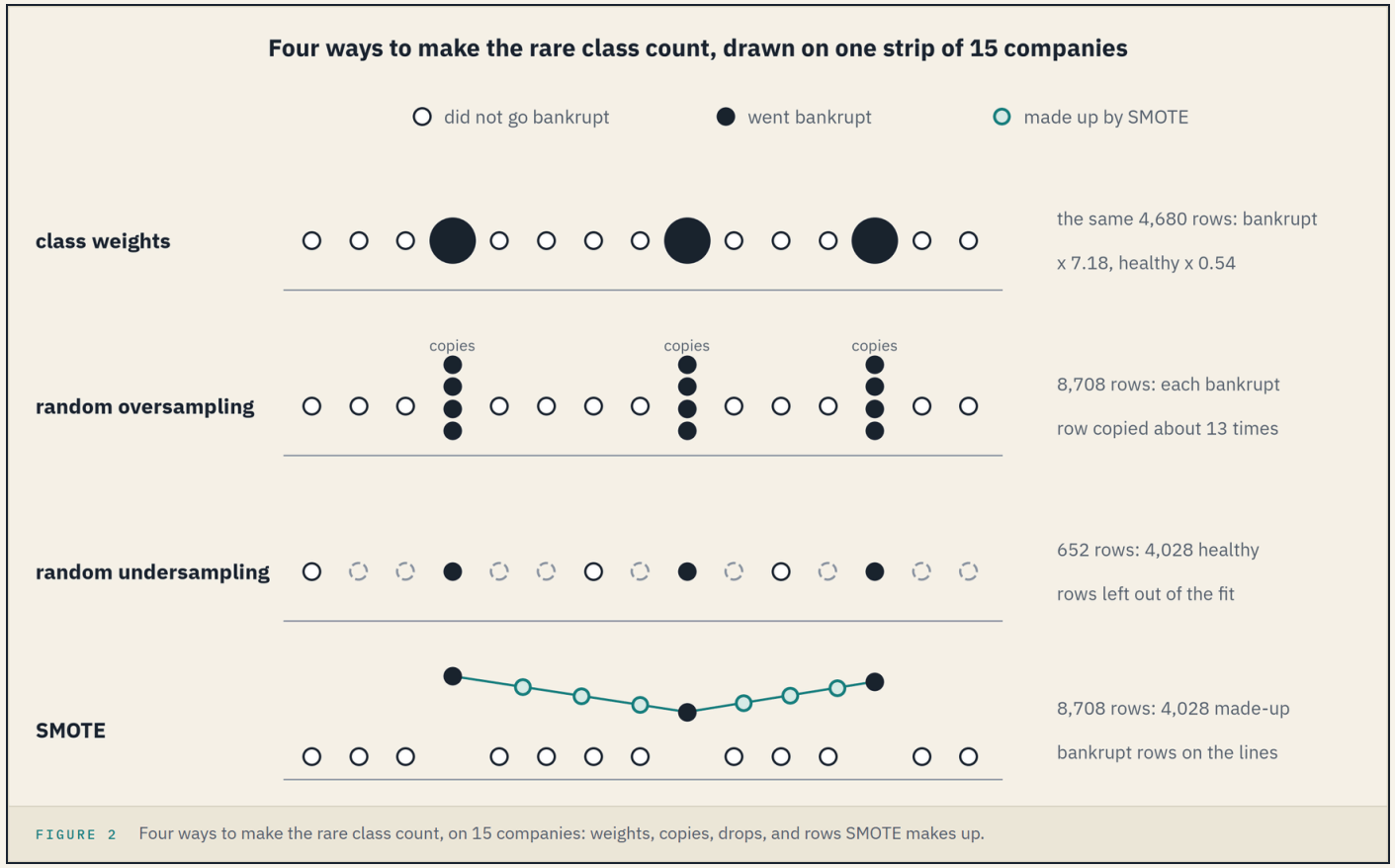
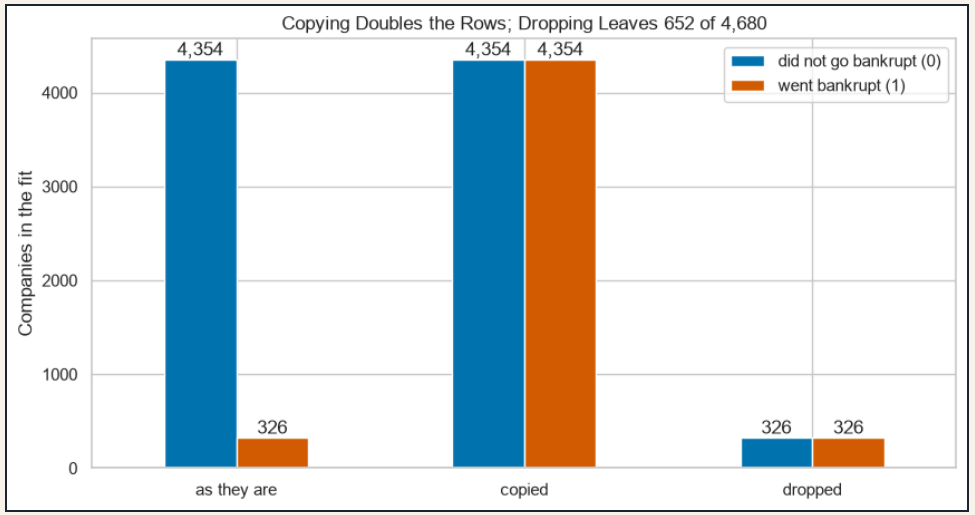


FIGURE 2 Four ways to make the rare class count, on 15 companies: weights, copies, drops, and rows SMOTE makes up.



ONE MADE-UP COMPANY

```
new_row = x + u * (neighbour - x)
```

That is SMOTE, synthetic minority oversampling: it repeats this step until the bankrupt companies match the healthy ones.

Resampling Inside the Folds

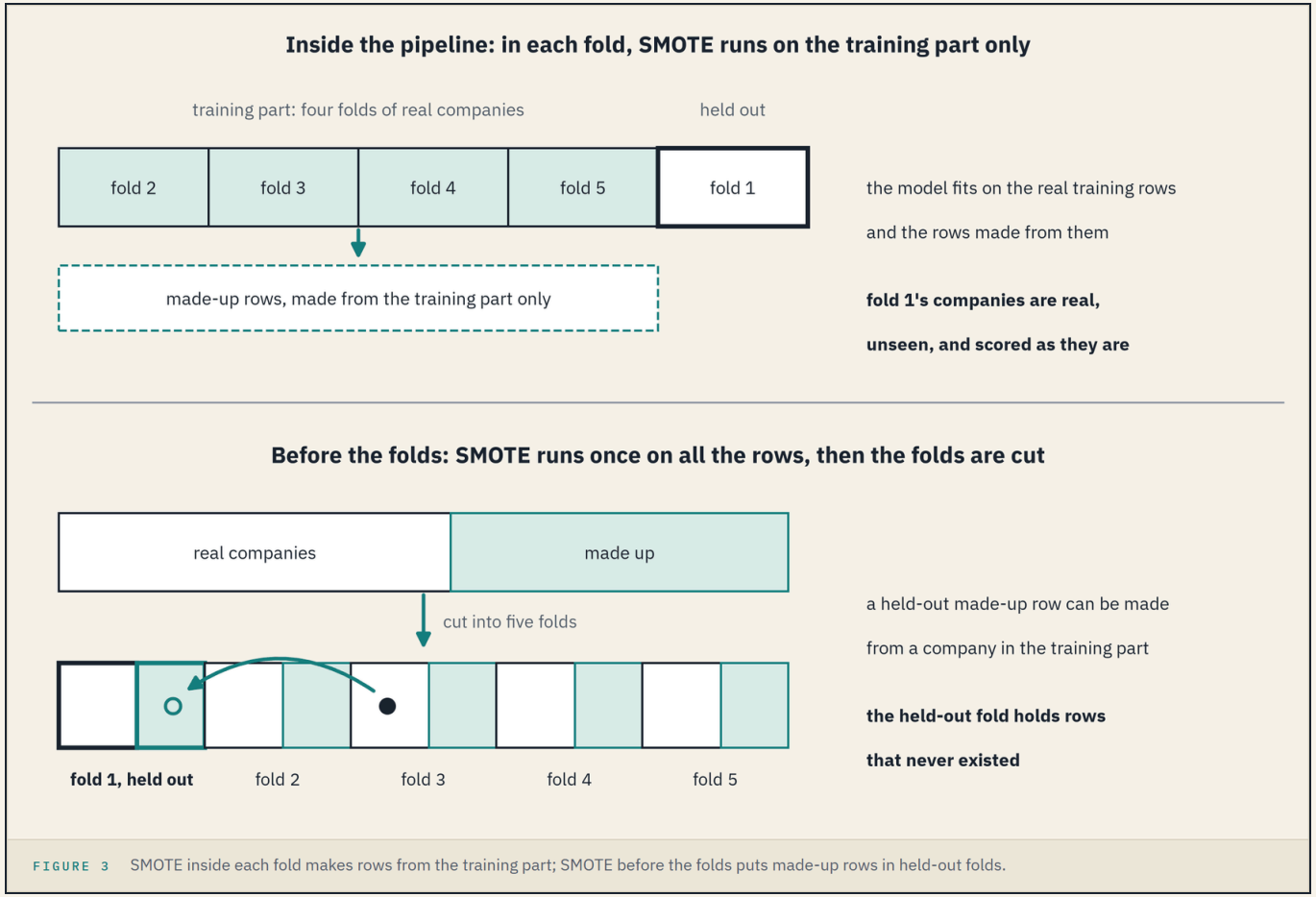


FIGURE 3 SMOTE inside each fold makes rows from the training part; SMOTE before the folds puts made-up rows in held-out folds.

THE QUESTION

Q. Run SMOTE once on all 4,680 rows, then score the same booster on five folds of the result. Write down its fold average precision.

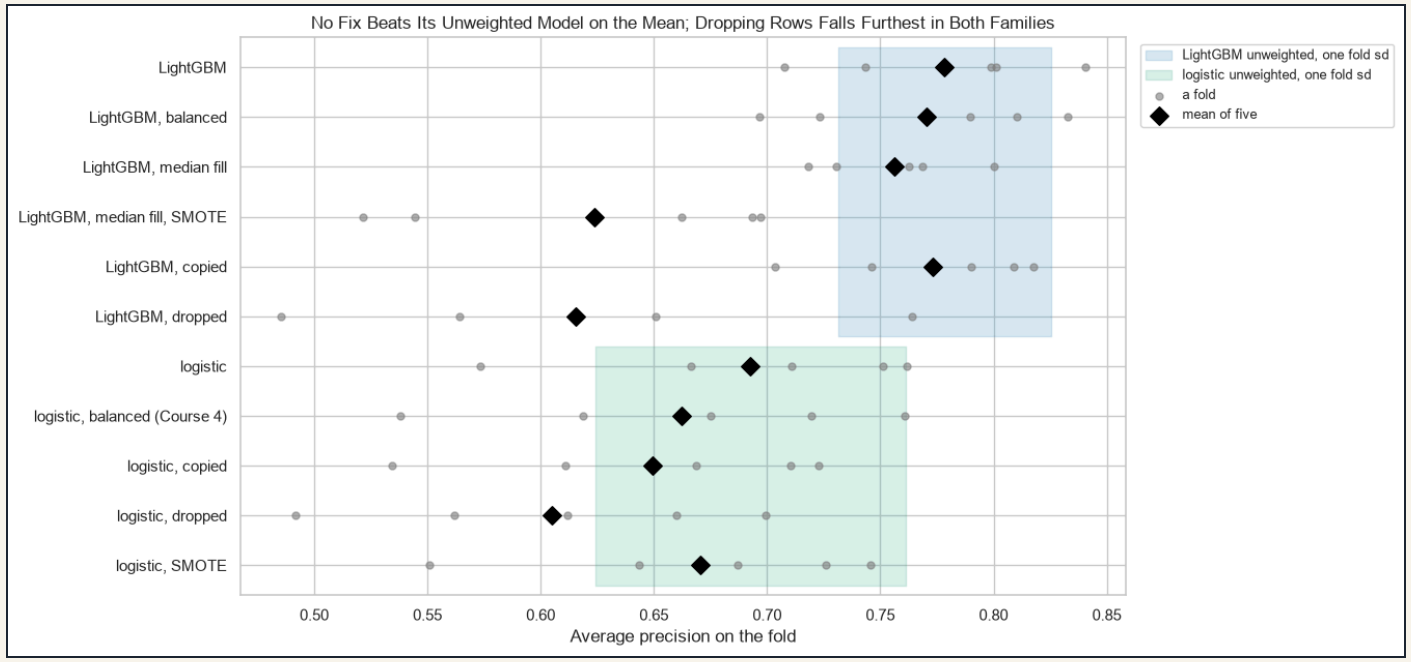
WHERE A SAMPLER RUNS

In an **`ImbPipeline`** a sampler runs at fit only.

In each fold it makes rows from that fold's training part, and the held-out companies are scored as they are.

Compare eleven models on folds and recall.

Model Performance Comparison



THE QUESTION

Q. How many companies must the median-filled booster with SMOTE flag to catch the same 229? Write down a number.

DAY 10'S RULE

Day 10's rule reads the highest threshold whose recall on the out-of-fold outputs is at least 0.70.

Weights, copies and the logistic with SMOTE stay inside one fold sd of it; dropping rows, and SMOTE on the booster, fall outside.

Add up each model's outputs against 326.

What Each Fix Does to the Output

THE QUESTION

Q. Course 4's balanced logistic gives each of the 4,680 companies an output. What do they add up to? Write down a number.

EACH MODEL'S MEAN OUTPUT, SUM AND BRIER SCORE

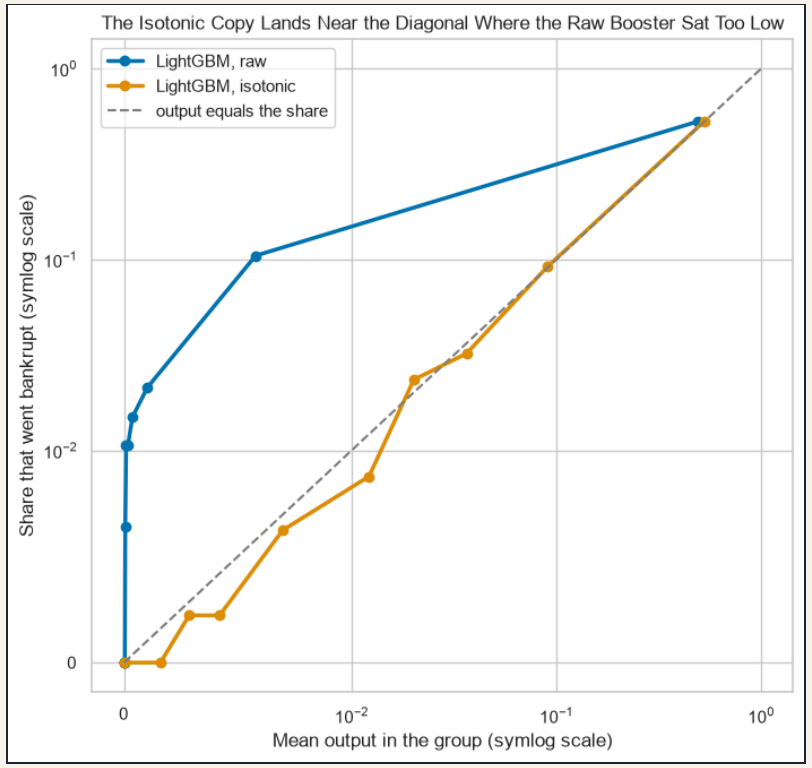
```
added = pd.DataFrame({"mean": outputs.mean(), "sum": outputs.sum(),
                      "Brier": [brier_score_loss(y_train, outputs[name]) for name in outputs]})
added
```

THE RUNNING SUM OF OUTPUTS AGAINST THE RUNNING COUNT OF BANKRUPTCIES

```
plt.figure(figsize=(9, 8))
for name in ["LightGBM", "LightGBM, median fill, SMOTE", "logistic", "logistic, balanced (Course 4)"]:
    order = np.argsort(-outputs[name].to_numpy())
    plt.plot(np.cumsum(went[order]), np.cumsum(outputs[name].to_numpy()[order]), linewidth=2.5, label=name)
```

The Brier score is the mean of (output - outcome) squared over the companies: it grows when outputs sit too high or too low, not only when the order is wrong.

Recalibration



THE QUESTION

Q. Sketch the corrected probability against the score, both from 0 to 1. Where does a score of 0.5 land?

THE PRIOR CORRECTION

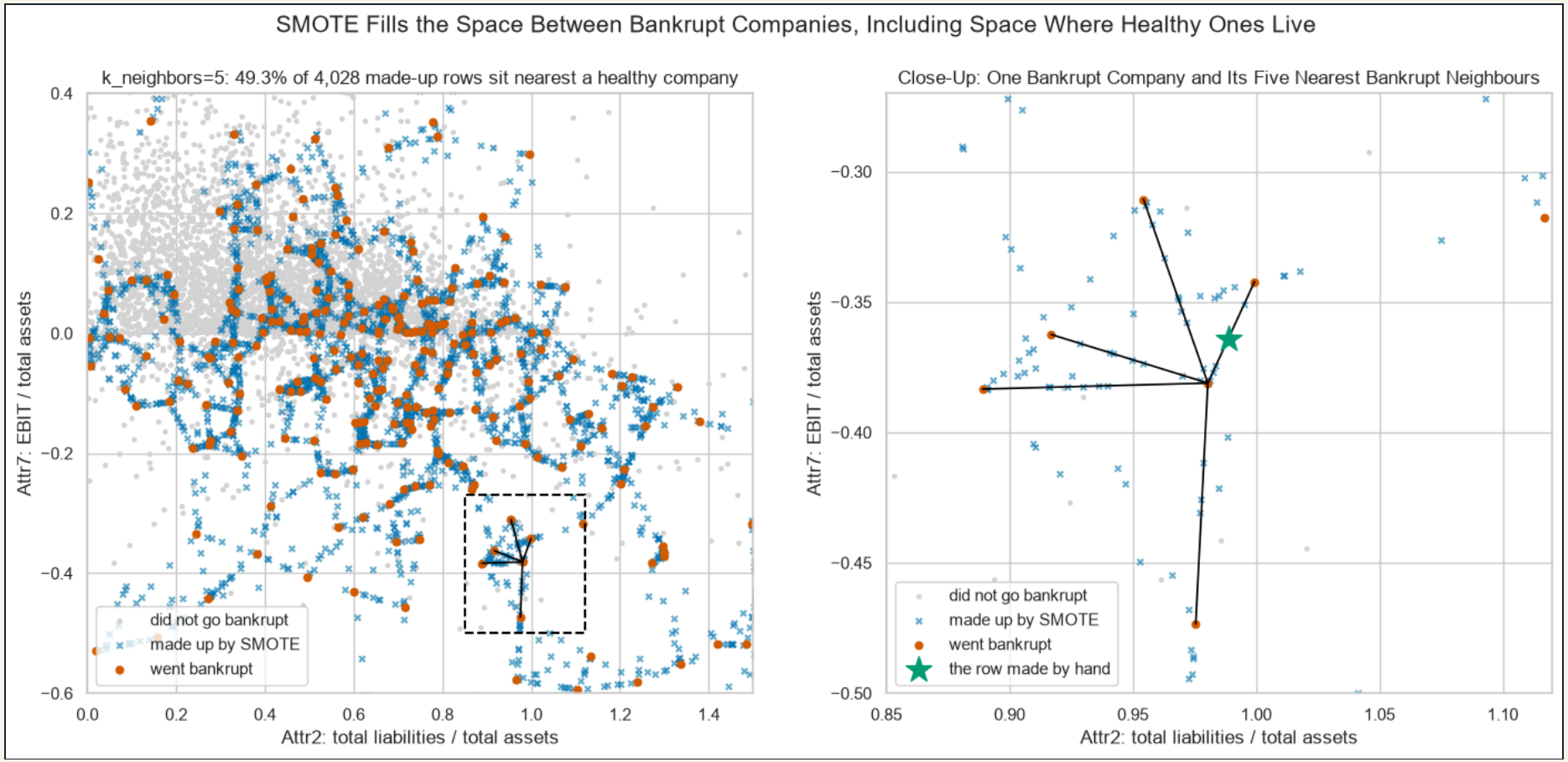
```
share = y_train.mean()
def prior_correction(score, fitted_share=0.5):
    odds = score / (1 - score) * (share / (1 - share)) / (fitted_share / (1 - fitted_share))
    return odds / (1 + odds)
```

ISOTONIC CALIBRATION

Isotonic calibration fits a staircase that only rises, from a model's output to the share that went bankrupt, on outputs from rows the model did not fit.

The unweighted LightGBM ranks; its isotonic copy reports a probability whose sum matches the rows.

Insights and Conclusion



Day 14 opens the unweighted LightGBM and splits its log-odds across the 64 ratios.

You Can Now

- Weight the rare class with **`class_weight="balanced"`** and say what the weights are.
- Copy, drop and synthesize rows with imbalanced-learn's samplers inside an **`ImbPipeline`**.
- Build one SMOTE row by hand from a company and a neighbour.
- Compare models on the same folds and at one recall, from out-of-fold outputs.
- Measure what a fix does to a model's output with its sum and the Brier score.
- Recalibrate a score with the prior correction and with isotonic calibration.

NEXT
WEEK 3 · DAY 4

SHAP Values

Day 14 opens the unweighted LightGBM and splits its log-odds across the 64 ratios.